

**Project Design Phase-II  
Technology Stack (Architecture & Stack)**

|               |                                                                                     |
|---------------|-------------------------------------------------------------------------------------|
| Date          | 28 June 2026                                                                        |
| Team ID       |                                                                                     |
| Project Name  | <b>Plugging into the Future: An Exploration of Electricity Consumption Patterns</b> |
| Maximum Marks | 4 Marks                                                                             |

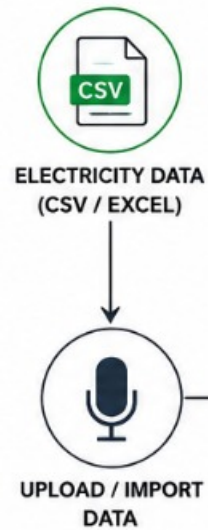
**Technical Architecture:**

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

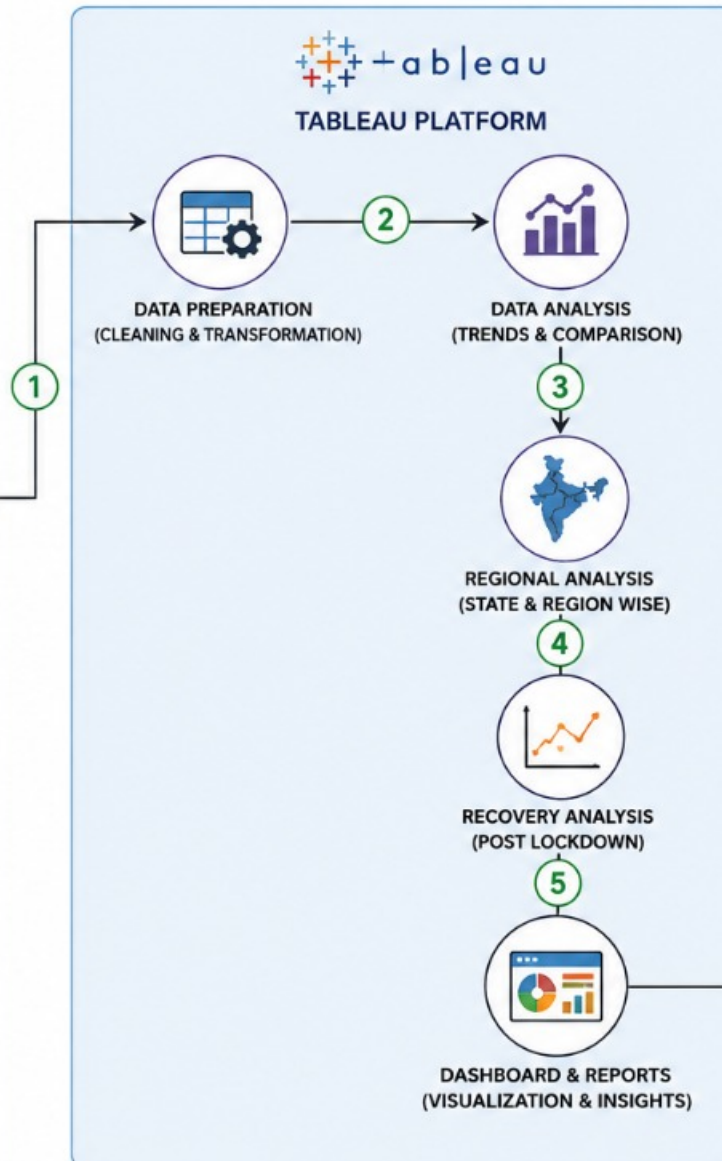
**Example: Order processing during pandemics for offline mode**

**Reference:** <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

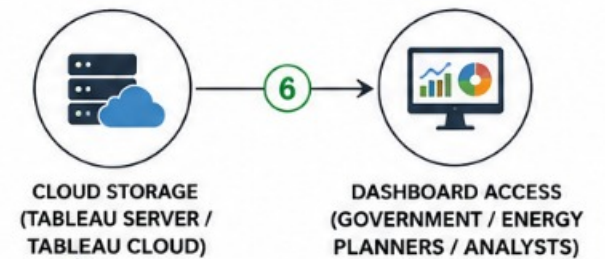
## USER



## SYSTEM (ANALYTICS PLATFORM)



## ADMIN / STAKEHOLDERS



- ① Import Electricity Data
- ② Analyze Trends & Compare (2019 vs 2020)
- ③ Regional Analysis (State / Region wise)
- ④ Post Lockdown Recovery Analysis
- ⑤ Generate Dashboard & Reports
- ⑥ Share & Access Dashboards

**Table-1 : Components & Technologies:**

| S.No | Component        | Description                                           | Technology           |
|------|------------------|-------------------------------------------------------|----------------------|
| 1    | Data Source      | Stores electricity consumption data for Indian states | CSV/Excel            |
| 2    | Data Preparation | Cleans and transforms the dataset                     | Microsoft Excel      |
| 3    | Data Analysis    | Performs trend and regional analysis                  | Tableau              |
| 4    | Visualization    | Creates charts, maps, and dashboards                  | Tableau              |
| 5    | Reporting        | Generates reports and insights                        | Tableau, MS Word     |
| 6    | Presentation     | Presents project findings                             | Microsoft PowerPoint |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description                                   | Technology             |
|------|-----------------|-----------------------------------------------|------------------------|
| 1.   | Platform        | Desktop-based analytics application           | Tableau Desktop        |
| 2.   | Data Type       | Structured data                               | CSV/Excel              |
| 3.   | User Interface  | Interactive dashboard with filters and charts | Tableau                |
| 4.   | Data Processing | Cleaning and preprocessing of data            | Microsoft Excel        |
| 5.   | Reporting       | Generation of analytical reports              | Tableau & MS Word      |
| 6.   | Accessibility   | Dashboard can be shared with stakeholders     | Tableau Public/Desktop |